

3. Requirements Gathering

This section defines the requirements for the Store Sales Time Series Forecasting project. It covers stakeholder needs, user stories, functional requirements, and non-functional criteria that the system must satisfy to deliver business value.

3.1 Stakeholder Analysis

Stakeholder analysis identifies the key individuals and groups affected by or involved in the project, along with their primary needs and level of influence.

Stakeholder	Role	Primary Need	Influence Level
Data Science Team	Project Executors	Accurate models, clean data pipelines, reproducible code	High
Academic Supervisor	Evaluator	Milestone delivery, documentation quality, methodology rigor	High
Corporación Favorita (simulated)	End Business	Reliable demand forecasts to reduce overstock and stockouts	Medium
Future Readers / Peers	Knowledge Consumers	Clear documentation, reusable code, well-explained methodology	Low

3.2 User Stories & Use Cases

User stories describe how different actors interact with the forecasting system to achieve specific goals.

ID	As a...	I want to...	So that...
US-01	Data Engineer	Load and merge all 6 dataset files into a single clean dataframe	The modeling pipeline has consistent, complete input data
US-02	EDA Analyst	Visualize sales trends by store, family, and date range	We can identify seasonal patterns and anomalies before modeling
US-03	ML Engineer	Generate lag and rolling features from historical sales data	The model captures temporal dependencies in the data
US-04	ML Engineer	Train and compare multiple forecasting models	We select the best-performing model for final submission

US-05	Project Manager	Track KPIs and milestone progress throughout the project	The team stays on schedule and within scope
US-06	Report Lead	Export model results and visualizations into Word reports	Milestones are documented and submitted on time

Key Use Case — Sales Forecasting Pipeline:

1. User uploads raw CSVs → system merges and validates data
2. System runs EDA and generates insight visualizations
3. Feature engineering pipeline transforms raw data into model-ready features
4. Model is trained, validated via time-series cross-validation, and evaluated
5. Predictions exported as submission CSV

3.3 Functional Requirements

Functional requirements define what the system must do — the specific behaviors, capabilities, and features the project must deliver.

ID	Category	Requirement	Priority
FR-01	Data	System must load and merge train.csv, stores.csv, oil.csv, holidays_events.csv, and transactions.csv on the date and store_nbr keys	Must Have
FR-02	Data	System must handle ~1,218 missing oil price values using forward-fill imputation	Must Have
FR-03	Data	System must flag and handle zero-sales entries for stores during their opening period	Should Have
FR-04	EDA	System must produce sales trend charts by store, product family, and time period	Must Have
FR-05	EDA	System must quantify the impact of holidays and promotions on sales using statistical analysis	Should Have
FR-06	Features	System must generate lag features (1, 7, 14, 28 days) and rolling mean/std features (7, 14 days) per store-family group	Must Have
FR-07	Features	System must encode temporal features: day of week, month, week of year, payday flag, and earthquake dummy (April 2016)	Must Have
FR-08	Modeling	System must train at least one baseline model and one advanced gradient boosting model (LightGBM or XGBoost)	Must Have
FR-09	Modeling	System must evaluate all models using time-series cross-validation and report RMSLE per fold	Must Have

FR-10	Output	System must generate a valid submission.csv format (id, sales columns)	Must Have
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Priority Legend: **Must Have** = Critical for project success **Should Have** = Important but not blocking

3.4 Non-Functional Requirements

Non-functional requirements define the quality attributes the system must exhibit — how well it performs its functions rather than what it does.

ID	Category	Attribute	Requirement	Metric / Target
NFR-01	Performance	Model Training Speed	Full LightGBM training on 3M rows must complete within a reasonable time	< 30 minutes per training run
NFR-02	Performance	Prediction Accuracy	The final model must achieve the target RMSLE score	RMSLE $\leq$ 0.45
NFR-03	Security	Data Privacy	All project data to be used only within authorized team members — no external sharing	Zero unauthorized access incidents
NFR-04	Security	Code Repository Access	GitHub repository must be private; access granted only to team members and supervisor	Private repo with role-based access
NFR-05	Usability	Code Readability	All notebooks must include section headers, inline comments, and a README explaining the pipeline steps	100% of notebooks documented
NFR-06	Usability	Reproducibility	Running notebooks top-to-bottom on a fresh environment must produce identical results using fixed random seeds	Seed fixed; results deterministic
NFR-07	Reliability	Pipeline Stability	The data pipeline must handle all 6 input files without runtime errors; all exceptions must be caught and logged	0 unhandled pipeline crashes
NFR-08	Reliability	Version Control	All code changes must be committed with descriptive messages; no direct pushes to the main branch	Git history maintained throughout
NFR-09	Maintainability	Modular Code Structure	Feature engineering and model training logic must be separated into distinct,	At least 4 separate modules

			importable modules or notebooks	
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